

RITARSHI ROY

B.Tech CS (Data Science) | NMIMS Mumbai

Mumbai, Maharashtra, IN

github.com/horserider2146

EDUCATION

Qualification	Institute	Board/University	Year	% / CGPA
MBA.Tech Data Science	Mukesh Patel School of Technology Mgmt & Engg (NMIMS)	NMIMS Mumbai	Currently Pursuing	8.46
XII	Delhi Public School, Ruby Park, Kolkata	CBSE	2024	75%
X	South Point High School, Kolkata	CBSE	2022	88%

CORE SKILLS

Languages:	Python, C, C++, SQL, Kotlin, JavaScript
Frameworks & Tools:	FastAPI, Express.js, React, Streamlit, Node.js, Retrofit, Room
Android Dev:	Kotlin, MVVM, LiveData, Coroutines, Material Design 3, MPAndroidChart
Databases:	SQLite, PostgreSQL, Redis, Room (Android)
AI / ML:	Anthropic Claude API, XGBoost, moving average forecasting, multi-agent pipelines
Other:	WebSocket, REST APIs, Docker, Git, GitHub, Power BI, SVG

PROJECTS

NetSentinel — Real-Time Network Security Simulator 2025

- Built an interactive network security simulator using Node.js, Express, and WebSockets with real-time SVG visualization
- Simulated 4 attack types: DDoS, SQL Injection, Port Scanning, and Man-in-the-Middle attacks
- Implemented layered defenses: per-node Firewall, IDS, IPS, Encryption, and global IP Blacklist
- Features a live Risk Analysis Dashboard, event timeline, and 3 scenario presets (Corporate, Cloud, IoT)
- Tech: Node.js, Express, WebSocket, Vanilla JS, SVG | github.com/horserider2146/NetSentinel

Relationship Intelligence System — AI Agent Pipeline 2025

- Designed a multi-agent AI pipeline (INA → PRA → ADA → RHSA → IRA → UAG → EBA) for relationship analytics
- Enforced Human-in-the-Loop (HITL) approval gate with cryptographic token validation before any action dispatch
- Built RESTful backend with FastAPI (Python) and React frontend; integrated Anthropic Claude API as LLM core
- Tech: Python, FastAPI, React, Anthropic API, Docker | github.com/horserider2146/relationship-intelligence-system

MarketLens — Crypto & Stock Trend Visualizer (Android) 2025

- Built a real-time cryptocurrency tracking Android app in Kotlin using CoinGecko API (no API key required)
- Implemented 7-day trend-based forecasting using moving average analysis; visualized with MPAndroidChart
- Features live search, Room-based watchlist, smart API caching, and Material Design 3 dark-themed UI
- Tech: Kotlin, MVVM, Room, Retrofit, Coroutines, MPAndroidChart | github.com/horserider2146/MarketLens

StochClaim — Insurance Fraud Detection Platform(Contributor) 2025

- Built an end-to-end insurance fraud detection platform using a five-layer stochastic AI pipeline from data ingestion to interactive dashboard
- Integrated Hidden Markov Models, gradient boosting, and probabilistic modelling for claim routing into approval, investigation, or denial categories

- LLM-powered natural language explanations via Groq API for interpretable fraud decisions; includes human review queues and Markov state transitions
- Tech: Python, FastAPI, Streamlit, Hidden Markov Models, XGBoost, Groq API
- Github: <https://github.com/addz9015/DataAnalysisAgent>

QuantumDDA — Multi-Table DBMS with Circuit Fingerprinting

2024

- Built end-to-end DBMS using SQLite, FastAPI, and Streamlit for quantum circuit data management
- Implemented circuit fingerprinting logic and an analytics dashboard for multi-table quantum circuit data
- Tech: Python, SQLite, FastAPI, Streamlit | github.com/horserider2146/quantum-circuit-dbms

Paperclip — Cross-Platform Clipboard Sharing Tool

2025

- Built a LAN clipboard sync tool between Android and Windows over TCP — no cloud or internet required
- Windows side: C#/.NET 8 system tray app using Win32 AddClipboardFormatListener to detect and broadcast clipboard changes
- Android side: Kotlin foreground service with Accessibility Service for background clipboard monitoring
- Protocol: newline-delimited JSON over persistent TCP socket with ping/pong and optional auth token support
- Tech: C#, .NET 8, Kotlin, TCP Sockets, Android Foreground Services, Win32 API

F1 Analytics Dashboard

2024 – Present

- Full-stack Formula 1 data platform integrating FastAPI, Streamlit, FastF1, PostgreSQL, Redis, and XGBoost
- Proposed and implemented a novel Driver Consistency Index metric for objective race performance evaluation
- Extended with 6 statistical analysis modules covering ANOVA, regression, logistic regression, and nonparametric tests across 18 new API endpoints
- Explored an emotion classification pipeline for F1 driver radio audio using ML
- Tech: Python, FastAPI, Streamlit, FastF1, PostgreSQL, Redis, XGBoost

INTERNSHIP EXPERIENCE

Summer Research Paper — Financial Market Crash Prediction

August 2025

Department of Data Science, MPSTME, NMIMS Mumbai

- Co-authored research paper: Predicting Financial Market Crashes Using a Hybrid Framework of Regime Switching, Stochastic Simulation, and Econophysics
- Proposed a hybrid model integrating Markov Regime Switching Models (MRS), Monte Carlo simulations, and Mean-Field Theory to predict financial crises
- Applied to two case studies: COVID-19 global market crash (2020) and Argentina 2001 crisis — successfully identifying regime transitions and critical temperature thresholds (Tc)
- Demonstrated that well-timed financial interventions near regime shift boundaries significantly influence long-term market stability
- Presented at the International Research Conference on Global Trends in Management and Innovative Strategies in Business, SVKM Usha Pravin Gandhi College of Arts, Science and Commerce — 12 August 2025

PROFESSIONAL ATTRIBUTES / ACHIEVEMENTS

- Built NetSentinel — a full real-time network security simulator with 4 attack types, layered defenses, and live risk dashboards
- Designed a multi-agent AI pipeline with cryptographic HITL enforcement using the Anthropic Claude API
- Shipped a published Android app (MarketLens v1.0) with live crypto data, 7-day ML forecasting, and a Room watchlist
- Built end-to-end full-stack projects independently spanning backend, database, frontend, and mobile layers
- Completed structured Power BI curriculum (Stages 1–5) with custom multi-table datasets
- Actively preparing for tech internships through DSA practice and open-source project development